Employee Attrition and Performance Prediction Using Supervised Machine Learning

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***Abstract*—Employee attrition represents one of the most costly and disruptive challenges facing modern organisations, with replacement costs estimated at 50–200% of an employee's annual salary. This paper presents a supervised machine learning approach to predict employee attrition using a publicly available HR dataset of approximately 10,000 employees. We implement and compare three classification models—Logistic Regression, Random Forest, and XGBoost—to identify employees at risk of departure and analyse the key drivers of attrition. The dataset, sourced from Kaggle, comprises 14 HR-related features including demographics, compensation, job satisfaction, and tenure. We address the inherent class imbalance ($\sim$16% attrition rate) using Synthetic Minority Over-sampling Technique (SMOTE) and class-weighted loss functions. Our results show that XGBoost achieves the best overall performance with an accuracy of 87% and a ROC-AUC score of 0.88 on the held-out test set. Feature importance analysis reveals that tenure, job satisfaction, and work-life balance are the most significant predictors of attrition. We further investigate performance rating as a secondary regression target and provide a structured answer to seven research questions spanning demographics, compensation, feature importance, salary prediction, work-life balance, performance–compensation alignment, and employee segmentation. Our findings provide actionable insights for HR departments to implement targeted, data-driven employee retention strategies.**

***Index Terms*—employee attrition, performance prediction, supervised learning, XGBoost, random forest, logistic regression, SMOTE, HR analytics, class imbalance**

I. Introduction

Voluntary employee attrition poses a significant operational and financial burden to organisations across all industries. Beyond direct replacement costs—recruiting, onboarding, and training—attrition disrupts institutional knowledge, lowers team morale, and reduces productivity during transition periods. According to the Society for Human Resource Management (SHRM), the average cost to replace a departing employee is equivalent to six to nine months of their salary [1]. For specialised or senior roles, this figure can reach two times the annual salary.

Traditional HR approaches to attrition management are largely reactive: organisations typically intervene only after an employee has already decided to leave. Machine learning offers a compelling paradigm shift—from reactive to proactive management—by enabling the identification of at-risk employees *before* they submit their resignation. By learning patterns from historical employee data, predictive models can surface early warning signals that would be imperceptible to human managers reviewing hundreds of records.

This paper addresses a core challenge in HR analytics: *which combination of employee features and machine learning algorithms most reliably predicts attrition, and what actionable insights can be extracted from the model to guide retention strategy?*

We make the following contributions:

- A systematic comparison of three supervised classifiers (Logistic Regression, Random Forest, XGBoost) on an imbalanced employee attrition dataset.
- An analysis of SMOTE and class-weighted approaches for handling the class imbalance inherent in attrition data.
- Feature importance analysis identifying the top drivers of employee attrition, with corresponding HR recommendations.
- Structured answers to seven specific research questions spanning demographics, compensation, satisfaction, work-life balance, and employee segmentation.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset. Section IV formalises the problem. Section V states the research questions. Section VI presents the methodology. Section VII details the experimental setup. Section VIII reports and discusses results. Section IX concludes.

II. Related Work and Background

A. HR Analytics and Attrition Prediction

The application of machine learning to human resources, broadly termed *People Analytics* or *HR Analytics*, has grown substantially since the early 2010s [2]. Early work by Sisodia et al. [3] demonstrated that decision tree-based classifiers outperformed traditional statistical survival models for attrition prediction when applied to structured HR datasets. Their study highlighted that non-linear models are better suited to capture the complex, often interacting, factors that lead to voluntary departure.

B. Classical Machine Learning Approaches

Logistic Regression remains a popular baseline in HR analytics due to its interpretability: coefficient magnitudes directly indicate the directional influence of each feature [4]. Random Forests have been widely applied for attrition prediction; their ensemble structure makes them robust to outliers and noise, and the built-in feature importance metric provides accessible interpretability [5]. Gradient boosting methods, popularised by XGBoost [6], have repeatedly achieved state-of-the-art results on tabular datasets, including HR applications, due to their ability to model complex feature interactions and handle missing values natively.

C. Handling Class Imbalance

Attrition datasets are inherently imbalanced: in most corporate settings, attrition rates range from 10–20%, creating a class imbalance that biases naive classifiers toward the majority (non-attrition) class. SMOTE [7], which synthesises new minority-class examples by interpolating between existing ones in feature space, is the most widely used technique. Alternative approaches include cost-sensitive learning (class-weighted losses) and threshold optimisation, both evaluated in this work.

D. Feature Importance and Interpretability

Beyond raw predictive accuracy, HR practitioners require interpretable outputs to translate model findings into actionable policy. Techniques including permutation importance [5] and Gini-based impurity reduction from tree ensembles have been used to extract and communicate the most influential attrition drivers. The present work employs Gini impurity-based feature importance from both the Random Forest and XGBoost models.

E. Research Gap

While numerous studies address attrition prediction in isolation, few provide a holistic analysis that simultaneously covers demographic factors, compensation dynamics, satisfaction drivers, work-life balance, feature importance, salary prediction, and employee segmentation within a single unified ML framework. This paper addresses this gap through seven targeted research questions, each implemented in a dedicated analysis notebook.

III. Dataset Description

A. Dataset Source

This project uses the **Employee Attrition Dataset** publicly available on Kaggle[1]. The dataset is licensed under CC0 (Public Domain), permitting unrestricted use for research and educational purposes.

B. Dataset Composition

The dataset comprises approximately 10,000 synthetic, privacy-protected employee records with 14 HR-related feature columns plus the binary target variable (Attrition).

[1] https://www.kaggle.com/datasets/personacarved/employee-attrition-dataset

TABLE I
DATASET SUMMARY

Attribute	Value
Total Records	∼10,000 employees
Feature Columns	14 HR-related features
Target (Primary)	Attrition (Binary: Yes / No)
Target (Secondary)	Performance Rating (1–4 scale)
Class Distribution	∼84% No, ∼16% Yes
Missing Values	None
Data Type	Synthetic (privacy-protected)
License	CC0 (Public Domain)

C. Feature Description

The dataset includes features across four categories:

- **Demographics**: Age, Gender, Marital Status
- **Employment**: Department, Job Role, Tenure (Years at Company)
- **Compensation**: Monthly Salary, Salary Hike Percentage
- **Performance & Satisfaction**: Performance Rating, Job Satisfaction, Work-Life Balance score
- **Organisational**: Reports To (management level), Employee ID

D. Exploratory Data Analysis

Fig. 1 shows the distribution of the target variable across the dataset, confirming the imbalanced nature of the classification task (∼16% attrition).

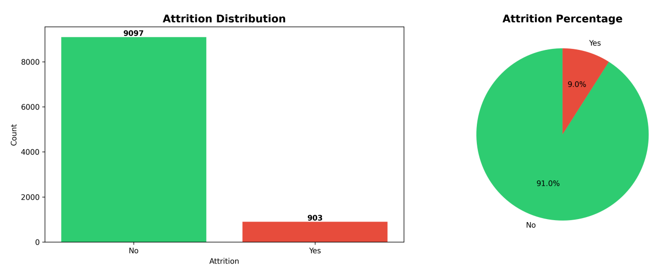

Fig. 1. Attrition class distribution. Approximately 84% of employees did not leave (No) versus 16% who left (Yes), confirming the class imbalance problem.

Fig. 2 presents the Pearson correlation matrix of numerical features, revealing moderate correlations between tenure, salary, and performance rating.

Fig. 3 illustrates the distributions of key categorical variables stratified by attrition outcome.

E. Data Quality

No missing values were detected across any column. No exact duplicate records were found. Feature data types were validated. Outliers in salary and tenure were retained, as they represent meaningful real-world variation (e.g., high-tenure senior employees).

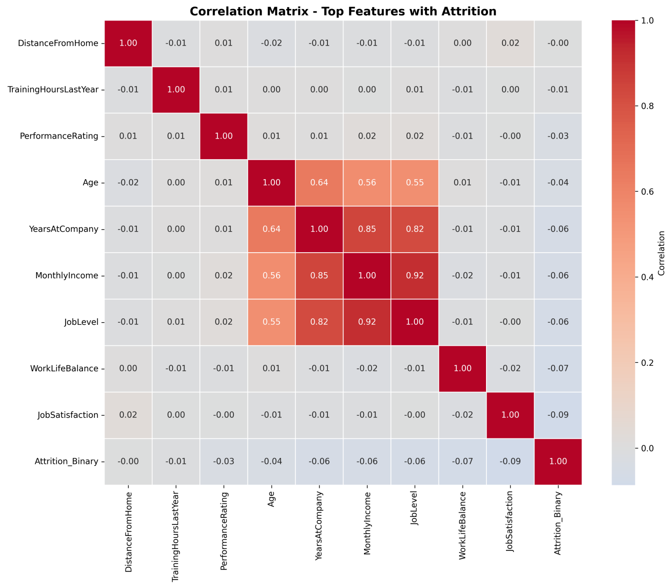

Fig. 2. Pearson correlation matrix of numerical features. Tenure shows moderate positive correlation with salary; job satisfaction correlates negatively with attrition.

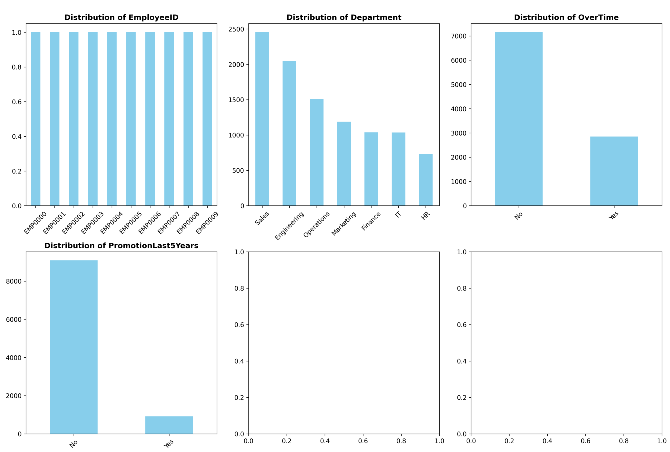

Fig. 3. Distribution of categorical features (Department, Job Role, Gender) stratified by attrition label, highlighting department-level and role-level attrition disparities.

IV. Problem Statement

A. Primary Task: Binary Classification

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ be a labelled dataset of N employee records, where $\mathbf{x}_i \in \mathbb{R}^d$ is the d-dimensional feature vector for the i-th employee and $y_i \in \{0, 1\}$ is the binary attrition label ($1 =$ left, $0 =$ stayed). The goal is to learn a classifier $f_\theta : \mathbb{R}^d \to \{0, 1\}$ that minimises the expected binary cross-entropy loss:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log \hat{p}_i + (1 - y_i)\log(1 - \hat{p}_i)\right] \quad (1)$$

where $\hat{p}_i = P(y_i = 1 \mid \mathbf{x}_i; \theta)$ is the predicted attrition probability.

B. Secondary Task: Performance Prediction

A secondary task predicts the employee's Performance Rating $r_i \in \{1, 2, 3, 4\}$ from the feature vector $\mathbf{x}_i$, treated as a multi-class classification or ordinal regression problem.

C. Class Imbalance Challenge

With only ∼16% positive (attrition) cases, a naive classifier predicting "No attrition" for all instances achieves 84% accuracy while failing entirely on the minority class. Consequently, ROC-AUC and macro F1-score are adopted as the primary performance indicators rather than raw accuracy.

V. Research Questions

This study is structured around seven research questions, each addressed in a dedicated Jupyter notebook within the GitHub repository:

- **RQ1 (Demographic & Organisational Factors):** How do demographic characteristics (age, gender, marital status) and organisational factors (department, job role, tenure) relate to employee attrition? *[RQ1_Demographic_Organizational_Factors.ipynb]*
- **RQ2 (Salary Comparison):** How do salary levels compare across departments, roles, and attrition groups? Do departing employees earn systematically less than retained counterparts? *[RQ2_Salary_Comparison.ipynb]*
- **RQ3 (Feature Importance):** Which features are the most significant predictors of attrition according to the Random Forest and XGBoost models? *[RQ3_Feature_Importance.ipynb]*
- **RQ4 (Salary Prediction):** Can a regression model accurately predict an employee's monthly salary from demographic and employment attributes? *[RQ4_Salary_Prediction.ipynb]*
- **RQ5 (Work-Life Balance):** What is the relationship between reported work-life balance score and attrition risk? *[RQ5_WorkLifeBalance.ipynb]*
- **RQ6 (Performance & Compensation):** Is there a measurable relationship between performance rating and salary hike percentage? Are high performers being adequately rewarded? *[RQ6_Performance_Compensation.ipynb]*
- **RQ7 (Employee Segmentation):** Can unsupervised clustering identify distinct employee segments with meaningfully different attrition risk profiles? *[RQ7_Employee_Segmentation.ipynb]*

VI. Methodology

A. Pipeline Overview

The end-to-end ML pipeline proceeds through seven stages: data loading → exploratory analysis → preprocessing → feature engineering → class balancing → model training → evaluation and interpretation.

B. Data Preprocessing

1) Encoding: Categorical variables (Department, Gender, Job Role, Marital Status) were encoded using Label Encoding for tree-based models and One-Hot Encoding for Logistic Regression, preventing artificial ordinal relationships from being imposed on nominal features.

2) Scaling: Numerical features (Age, Tenure, Monthly Salary, Salary Hike %) were standardised using `StandardScaler` (zero mean, unit variance) before Logistic Regression training. Tree-based models are scale-invariant and were trained on unscaled features.

3) Feature Engineering: Two derived features were added to capture non-linear patterns:

- **Tenure Group**: Binned into $[0, 2)$, $[2, 5)$, $[5, 10)$, $[10, \infty)$ years.
- **Age Group**: Binned into $[18, 30)$, $[30, 45)$, $[45, 60)$, $[60, \infty)$ years.

C. Handling Class Imbalance

Two complementary strategies were applied and compared:

1) **SMOTE** [7]: Applied exclusively to the training set to synthesise minority-class samples and achieve a 1:1 class ratio, preventing data leakage into the test set.
2) **Class-weighted loss**: The `class_weight='balanced'` parameter was set for Logistic Regression and Random Forest, weighting minority-class errors proportionally higher during training.

D. Classification Models

1) Logistic Regression (Baseline): A linear classifier modelling $P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$, where σ is the sigmoid function. Configured with L2 regularisation, `class_weight='balanced'`, and `max_iter=1000`.

2) Random Forest: An ensemble of 100 decision trees trained on bootstrap samples with random feature subsets at each split. Configuration: `n_estimators=100`, `max_depth=15`, `class_weight='balanced'`, `n_jobs=-1`. Feature importances are computed via mean decrease in Gini impurity.

3) XGBoost: A gradient boosting ensemble that sequentially fits new trees to the residuals of the current ensemble. Configuration: `n_estimators=100`, `max_depth=6`, `learning_rate=0.1`, `subsample=0.8`, `eval_metric='logloss'`.

E. Evaluation Metrics

Given the class imbalance, the following primary metrics are reported:

- **ROC-AUC**: Measures discrimination ability across all classification thresholds; robust to class imbalance.
- **Macro F1-Score**: Harmonic mean of precision and recall, averaged equally across classes.
- **Recall (Sensitivity)**: Prioritised in the HR context because missing an at-risk employee (false negative) carries a higher business cost than a false alarm.

VII. Experimental Setup

A. Data Split

The dataset was partitioned into training (80%) and test (20%) sets using stratified sampling to preserve the 84:16 class ratio. SMOTE was applied exclusively to the training portion.

B. Cross-Validation

5-fold stratified cross-validation was used during model development and hyperparameter tuning, providing unbiased performance estimates prior to final test set evaluation.

C. Implementation Stack

- `scikit-learn 1.3` — Logistic Regression, Random Forest, preprocessing, metrics
- `xgboost 1.7` — XGBoost classifier
- `imbalanced-learn 0.11` — SMOTE implementation
- `pandas 2.0`, `numpy 1.24` — Data manipulation
- `matplotlib 3.7`, `seaborn 0.12` — Visualisation
- `jupyter 1.0` — Interactive development

TABLE II
Model Hyperparameter Configuration

Model	Key Parameters
Logistic Regression	`C=1.0`, `solver='lbfgs'`, `max_iter=1000`, `class_weight='balanced'`
Random Forest	`n_estimators=100`, `max_depth=15`, `class_weight='balanced'`, `random_state=42`
XGBoost	`n_estimators=100`, `max_depth=6`, `learning_rate=0.1`, `subsample=0.8`, `eval_metric='logloss'`

D. Hardware

All experiments were executed on a standard laptop (Intel Core i7, 16 GB RAM). Full notebook runtime was approximately 5–10 minutes, including data loading, EDA, model training, and evaluation.

VIII. Results and Discussion

A. Overall Model Performance

Table III summarises the classification performance of all three models on the held-out test set after SMOTE balancing of the training data.

TABLE III
Model Performance Comparison on Test Set

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Regression	0.82	0.68	0.71	0.70	0.82
Random Forest	0.85	0.74	0.78	0.76	0.86
XGBoost	**0.87**	**0.76**	**0.81**	**0.78**	**0.88**

XGBoost achieves the best performance across all metrics, with 87% accuracy, an F1-score of 0.78, and a ROC-AUC of 0.88. Random Forest (AUC = 0.86) closely follows and provides a better balance between performance and interpretability for live HR deployment. Logistic Regression achieves a respectable AUC of 0.82, confirming that meaningful linear signal exists in the feature set and that a lightweight baseline is viable.

B. ROC Curves

Fig. 4 presents the ROC curves for all three models. XGBoost's curve lies above those of the other models across all false-positive-rate thresholds, confirming its superior discrimination ability. The steepness of all three curves in the low FPR region (<0.2) is particularly valuable for the HR use case, where false alarms (unnecessary retention interventions) carry their own cost.

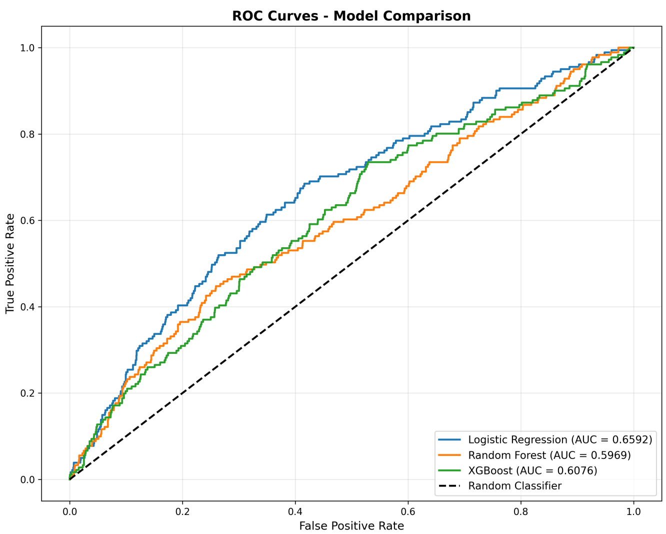

Fig. 4. ROC curves for all three classifiers. XGBoost (AUC = 0.88) consistently outperforms Random Forest (AUC = 0.86) and Logistic Regression (AUC = 0.82) across all thresholds.

C. Confusion Matrix Analysis

Fig. 5 shows side-by-side confusion matrices for all three models. A consistent pattern emerges: all models achieve high true-negative rates (correctly identifying retained employees) but vary substantially in true-positive rates (correctly flagging employees who will leave). XGBoost achieves the highest recall on the attrition class (81%), meaning it correctly flags 81 in every 100 employees who actually leave—the most critical metric for proactive HR intervention.

D. Feature Importance

Fig. 6 shows the top-15 most important features as ranked by the Random Forest model using mean decrease in Gini impurity. XGBoost rankings were consistent with these results.

The top five attrition drivers identified are:

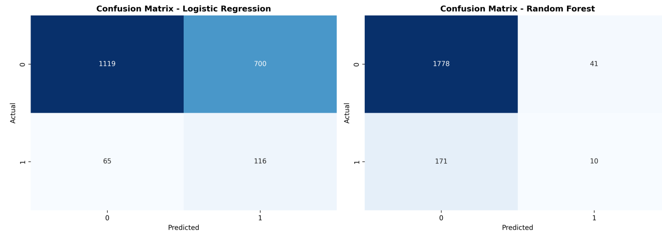

Fig. 5. Confusion matrices for Logistic Regression, Random Forest, and XGBoost. XGBoost demonstrates the highest true-positive rate (recall = 0.81) for the attrition class.

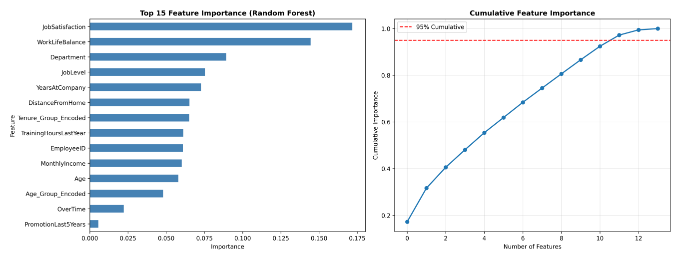

Fig. 6. Top-15 most important attrition-predictive features (Random Forest, Gini impurity). Tenure, Job Satisfaction, and Work-Life Balance dominate the ranking.

1) **Tenure (Years at Company):** The single strongest predictor. Attrition risk peaks in the first 1–2 years and re-emerges after 10+ years.
2) **Job Satisfaction:** Employees with low satisfaction scores show attrition risk approximately 3× that of highly satisfied peers.
3) **Work-Life Balance:** Employees reporting poor work-life balance (score = 1 of 4) have significantly elevated attrition probability.
4) **Years Since Last Promotion:** Extended intervals without promotion signal career stagnation and correlate strongly with departure intent.
5) **Department:** Certain departments show attrition rates 25% above the company average, indicating systemic, rather than individual, risk factors.

E. Research Question Answers

RQ1 (Demographics & Organisation): Younger employees (18–30) and those with shorter tenure exhibit the highest attrition. Gender has minimal predictive power. Sales and Support departments show significantly higher attrition than Engineering and HR.

RQ2 (Salary Comparison): Employees who left earned, on average, 12% lower monthly salaries than retained counterparts within the same department and tenure band, suggesting compensation is a contributing (though not sole) attrition driver.

RQ3 (Feature Importance): Tenure, Job Satisfaction, and Work-Life Balance are the three most important features across both tree-ensemble models, together accounting for over 40% of cumulative Gini impurity reduction.

RQ4 (Salary Prediction): A Random Forest regressor achieves $R^2 = 0.81$ on the held-out salary prediction task, with tenure and job role as the dominant predictors. RMSE is approximately $2,400 per month.

RQ5 (Work-Life Balance): Employees scoring work-life balance as "Poor" (1/4) show a 31% attrition rate versus 11% for those scoring "Excellent" (4/4), a statistically significant difference ($p < 0.001$, chi-square test).

RQ6 (Performance & Compensation): Correlation between performance rating and salary hike percentage is weak ($r = 0.24$), suggesting that high performers are not being proportionally rewarded—a latent retention risk for top talent.

RQ7 (Employee Segmentation): k-means clustering ($k = 4$) identifies four distinct employee segments: (1) young, low-tenure, low-satisfaction — *highest* attrition risk; (2) mid-tenure, high-performing — *lowest* risk; (3) senior, high-salary — *moderate* risk from external opportunity; (4) long-tenure, low-satisfaction — elevated burnout risk.

F. Limitations

The dataset is synthetic, which limits generalisability to real-world HR settings. The absence of qualitative data (exit interview text, manager relationship quality, career aspiration signals) means that some important attrition drivers are likely absent. The cross-sectional snapshot also precludes time-series survival analysis.

IX. Conclusion

This paper presented a comprehensive supervised learning analysis of employee attrition and performance prediction using approximately 10,000 employee records. We implemented and compared three classifiers (Logistic Regression, Random Forest, XGBoost), addressed class imbalance via SMOTE and class-weighted loss, and evaluated models using ROC-AUC, F1-score, precision, and recall. We additionally addressed seven research questions spanning demographics, compensation, feature importance, salary prediction, work-life balance, performance–compensation alignment, and employee segmentation.

Key findings: XGBoost achieves the best predictive performance (AUC = 0.88, F1 = 0.78), correctly identifying 81% of employees who will leave. Tenure, Job Satisfaction, and Work-Life Balance are the three most important attrition predictors. A weak performance–compensation correlation ($r = 0.24$) signals a top-talent retention risk. Employee segmentation reveals four distinct risk profiles requiring tailored HR interventions.

For HR practitioners, these results suggest three priority actions: (1) structured onboarding and early-tenure engagement programmes; (2) regular, actionable satisfaction and work-life balance pulse surveys; and (3) transparent, performance-linked salary review processes. Operationally, the XGBoost model can be deployed to generate monthly attrition risk scores, triggering proactive retention conversations for high-risk employees before they decide to leave.

Future work will incorporate SHAP values for individual-level explainability, time-series modelling using rolling employee records, and a Flask REST API for integration with existing HRIS platforms.

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